

ECE 501 – DIGITAL IMAGE PROCESSING

Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

Weekly Progress Report

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1. Introduction

During Weeks 2 and 3, the team studied the proposed Content-Based This is an organized AID multi-label aerial image retrieval system, called the Image Retrieval (CBIR) system. implemented visual-feature extraction and retrieval, and created image data set. in stages. The purpose of this is to get a visual and semantic recapture. Handcrafted images, color, texture, shape, and space to create similar images No machine learning, OpenCV or scikit-image is used.

2. Background Study

CBIR retrieves images by comparing the numerical features extracted from Make use of image content instead of relying on the name of the image. Since aerial images may have several semantic labels, retrieval is tested with both scene-class agreement and multi-label similarity.

The general pipeline is:

Image → Preprocessing → Feature Extraction → Similarity Ranking → Top- K

Pillow is used for image handling, NumPy for feature calculations, and Matplotlib for retrieval visualizations.

3. Dataset Organization

The data consists of 2400 gallery images and 600 query images. Images from 30 different scene classes. There are 17 images with each image being associated with each. Reads all the multi-label annotation information and returns the corresponding binary semantic labels. Images returned by the Gallery are precomputed and cached, and query images are Compared with the gallery. Compared to the gallery. Normalization statistics are calculated All photos are from gallery.

4. Methodology and Progress

Phase 0 — Dataset Preparation: The indexing utilities find images, link image identities and scenes. load semantic labels, and load classes. A split that is replicable With seed 42 selects 20 images from the gallery, discards: There are 1,800 images in this gallery, and 600 images in the validation gallery for this split. The A separate set of 600 images is used to evaluate the retrieval performance.

Phase 1 — Color-Based Retrieval: Histograms of RGB and HSV are used to report image-color distributions. The RGB The HSV descriptor has 4096 bins (16 cubed). Has $30 \times 32 = 960$ hue-saturation bins. L1-normalized Histogram intersection is used to compare histograms:

$$S(H_q, H_g) = \sum_i \min(H_q(i), H_g(i)).$$

A higher score indicates greater color similarity. RGB and HSV provide the initial retrieval baseline.

Phase 2 — Texture Descriptors: Three handcrafted descriptors capture local patterns and repeated structures. Local Binary Pattern (LBP) uses the values of the neighboring pixels around each pixel with its value. Make a 59-bin histogram for the eight neighbours. The Gray-Level Co-occurrence Matrix (GLCM) consists of 32 gray levels, Two distances, 4 orientations for the calculation of contrast, Generating dissimilarity, homogeneity, energy and correlation, 40 features. A Gabor filter bank is a set of four orientations and three sizes. A mean and standard deviation of 12 filter responses were computed for frequencies. produce 24 features. LBP employs histogram intersection. For standardized GLCM and Gabor features cosine similarity is used. The three scores are combined to provide an average texture similarity.

Phase 3 — Shape and Edge Descriptors: Images are resized to 128×128 grayscale. Histogram of In Oriented Gradients (HOG) there are 9 bins for orientation. The cell size is 16×16 cells and the normalization is L2-Hys. blocks, producing 1,764 features. A Sobel edge map is which is summarized as “normalized edge density” in a 4×4 grid, producing 16 features. Seven Hu invariant moments are computed From the edge map and sign preserving log-scaled.

Shape similarity is fused as:

$$S_{\text{shape}} = 0.60S_{\text{HOG}} + 0.25S_{\text{edge}} + 0.15S_{\text{Hu}}.$$

Color-texture-shape retrieval averages the RGB, combined-texture, and combined-shape similarity scores.

Phase 4 — Spatial-Layout Descriptors: Spatial features maintain coarse level of information associated with the visual properties occur. Grid Color Moments divide the 128×128 Load HSV image, divide into 4×4 cells and save the mean and standard. 96 features are obtained as the deviations of each HSV channel. Grid Intensity Normalized mean grayscale intensity in the same 16 cells measured by stores. The Color Layout Descriptor (CLD) is an average of the luminance in an 8×8 grid applies a 2 dimensional DCT and keeps the The low-frequency coefficients in the top-left 4×4 window, resulting in 16 features.

The combined spatial score is:

$$S_{\text{spatial}} = 0.60S_{\text{moments}} + 0.20S_{\text{intensity}} + 0.20S_{\text{CLD}}.$$

The four-cue retrieval score equally combines RGB color, Gabor texture, HOG shape, and Grid Color Moments:

$$S_{4\text{-cue}} = \frac{S_{\text{RGB}} + S_{\text{Gabor}} + S_{\text{HOG}} + S_{\text{moments}}}{4}.$$

5. Evaluation and Current Findings

The evaluation scripts consider the images in all 600 query images and rank them according to them. Generate and document results overall and by scene class. Relevance is assessed Agreement of labels within the same class and semantic-label Jaccard similarity. thresholds of 0.5 and 0.8. Precision at $k = 5$, Precision at $k = 10$, Hit at 5 and mean Average Precision (mAP). Retrieval visualizations For each of the 30 scene classes, a set of men are generated.

The provided reports have different summary values. The detailed Prior to checking the comparison tables and cumulative results CSV, it is advisable to review the following: final submission. The cumulative results report same-class Precision@5 of 42.03% for the color baseline, 48.57% for For color–texture fusion, the result is 50.97% and for color–texture–shape fusion, it is 52.50%. For the four-cue spatial fusion, the scores were 50.10% for both.

The results show that the texture and shape can increase the Retrieval accuracy was reported with respect to the color base line. The current Four-cue result is slightly lower than Phase 3 result, indicating that Adding another feature doesn't always mean better performance on all metrics.

6. Conclusion and Next Steps

Phases 0–4 set up a classical pipeline for CBIR with dataset. Indexing, colour, texture, shape and spatial descriptors. Cached combine results from multiple features to get a better score.by fusing results from different features using similarity scores. Integrates visual information. The following steps to take are: Check all evaluation outputs, look at the results per class. and failure cases, and complete individual The descriptors are against the fused system.